		The direction of F_n and g is from higher potential to lower potential, i.e. P to Q
12	**D**	For simple harmonic motion, $a = -\omega^2 x$. Hence the gradient of an a-x graph is $-\omega^2$ $\lvert\text{Gradient}\rvert = = (\frac{2\pi}{T})^2 = \frac{4\pi^2}{T^2}$ When the period of oscillation is doubled, the gradient of the graph is one quarter of the original graph.
13	**B**	$v = f\lambda \rightarrow 340 = (500)\lambda \rightarrow \lambda = 0.68$ m